Phase 1 - Grade 1 - Diagnostic

Key Learning Points

List

Ordered, MUTABLE collection; can hold mixed types; allows duplicates. Supports in-place changes e.g. my_list.append(4).

Tuple

Ordered, IMMUTABLE collection. Same as a list otherwise (mixed types, duplicates allowed). The ONLY real difference from a list is mutability -- once created, it cannot be changed. my_tuple.append(4) raises AttributeError. Used when data should never be accidentally mutated (e.g. a NumPy array's .shape is always a tuple).

List comprehension

[expression for item in iterable if condition] -- concise one-line way to build a list, equivalent to a for-loop with an optional filter.

range(n)

Generates integers from 0 up to but NOT including n. range(5) -> 0,1,2,3,4 (five values, stopping one short of the argument).

Clarified Doubts

My doubt: Can you explain how range() works?

Clarification: range(n) generates a sequence starting at 0, stopping ONE SHORT of n. range(5) gives 0,1,2,3,4 -- think of it as '5 numbers starting from 0'. To get 1 through 5 you'd write range(1,6).

Examples

```
def filterout_evennumber(number_data: list):
    even_num = [x for x in number_data if x % 2 == 0]
    return even_num

result = [x**2 for x in range(5) if x % 2 == 0]
# range(5) -> 0,1,2,3,4
# keep even x -> 0, 2, 4
# square each -> 0, 4, 16
# result = [0, 4, 16]
```

Quiz Q&A;

Q1. Write a function that filters a list down to only even numbers.

A: [x for x in data if x % 2 == 0]

Q2. What is the difference between a Python list and a tuple?

A: Mutability: a list can be changed after creation (append/remove/modify); a tuple cannot. Everything else (ordered, mixed types, duplicates allowed) is the same.

Q3. What does [x2 for x in range(5) if x % 2 == 0] evaluate to?**

A: [0, 4, 16] -- range(5) gives 0-4; even values are 0,2,4; squared gives 0,4,16.